

**UCSC****University of Colombo, Sri Lanka***University of Colombo School of Computing***DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**Academic Year 2023 — 3rd Year Examination — Semester 6**IT6206 — Software Quality Assurance***Structured Question Paper*

(2 Hours)

To be completed by the candidate**Index Number**

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Important Instructions

- The duration of the paper is **2 hours**.
- The medium of instructions and questions is English. Students should answer in the medium of English language only.
- This paper has **4 questions** on **12 pages**. Answer **all** questions.
- All questions carry **equal** marks.
- Write your answers **only on the space provided** on this question paper.
- Do not tear off any part of this question paper. Under no circumstances may this paper (or any part of this paper), used or unused, be removed from the Examination Hall by a candidate.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not allowed**.
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To be completed by the examiners

1	
2	
3	
4	
Total	

- 1) (a) What is meant by “Software Test Products” in software testing process. (04 marks)

Work products is the generic name given to any form of documentation, informal communication or artifact that is used in testing

- (b) There are seven categories of test work products listed in the answer box. Select the correct Software Test Product for each category using the numbered list given below. Write down the sequence number of the correct choice in the box provided in front of the category. (1x7 = 07 marks)

List of Test Work Products:

- | | | |
|---------------------------|-------------------|------------------------|
| 1 Test cases | 4 Test reports | 7 Test summary reports |
| 2 Test plans | 5 Defect reports | |
| 3 Test execution schedule | 6 Test conditions | |

Test planning work products	2
Test monitoring and control work products	4
Test analysis work products	6
Test design work products	1
Test implementation work products	3
Test execution work products	5
Test completion work product	7

(c) State whether the following statements are true (T) or false. (1x10 =10 marks)

1. Regression testing is done before code fixes, upgrades or any other system maintenance.
2. In clear-box testing, the code is visible to the testers.
3. Decision, branch and condition coverage are important Glass-box testing techniques.
4. Behavioural testing focuses on testing functionalities of software applications with the knowledge of internal code structure, implementation details and internal paths.
5. Finite State Machine Coverage is one of the Black-box testing techniques to ensure how many time-specific states are visited and transited.
6. Stress testing is measuring software on its robustness and error handling capabilities under extremely heavy load conditions.
7. Static testing is about the prevention of defects whereas dynamic testing is about finding and fixing the defects.
8. Test Scenarios Identification, Test Cases Writing and Test Data Preparation are activities in test case development phase.
9. Deferred, Occupied, and Initiated are some of the states of Bug lifecycle.
10. Integration testing reveals incorrect sequencing or timing of interface calls.

F

T

T

F

F

T

T

T

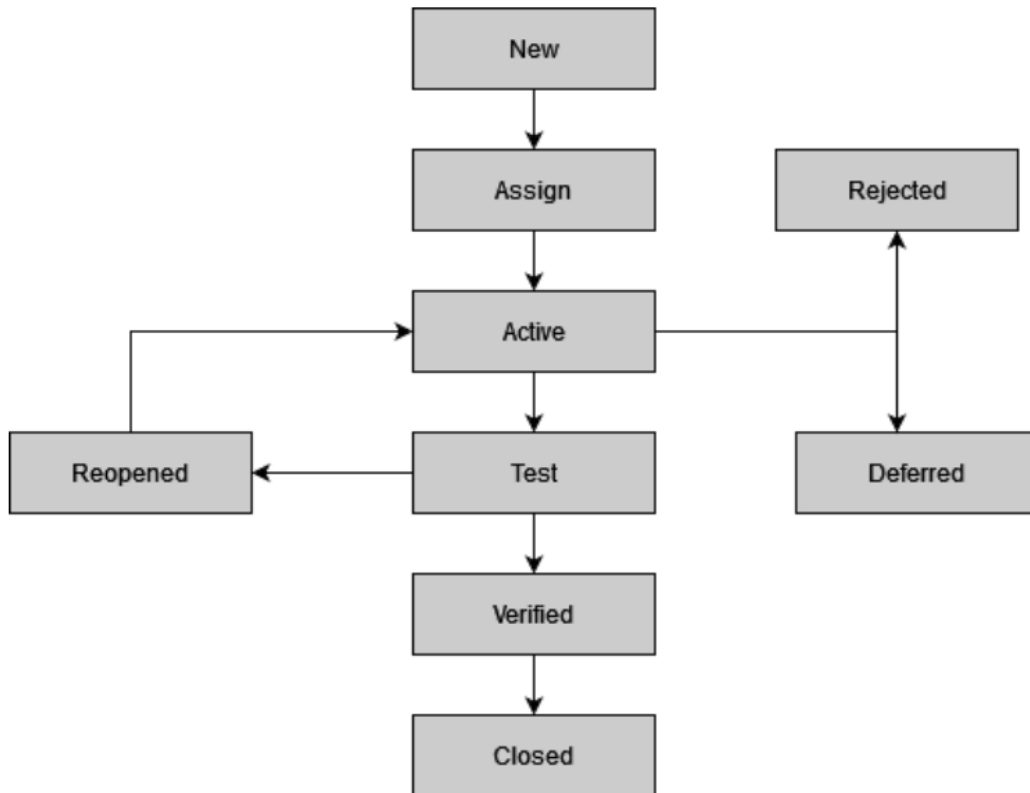
F

T

(d) List down four (4) black box testing types. (04 marks)

Equivalence Partitioning
Boundary Value Analysis
Decision Table Testing
State Transition Testing
Use Case Testing

- 2) (a) Defect Lifecycle shows the journey of a defect, the cycle which a defect goes through during its lifetime. Draw a suitable diagram to show the defect lifecycle states and their interactions. (10 marks)



- (b) List down the six phases of software testing lifecycle. (06 marks)

Requirement Analysis, Test Planning, Test Cases Designing, Test Environment setup, Test execution, Test closure

- (c) What is Software test automation? (04 marks)

Automation Testing is the use of software tools to develop and execute tests. Tools used in test automation can enter test data into the applications, compare expected and actual results and generate reports.

- (d) Give three advantages and two risks of software test automation. (05 marks)

Advantages:

Speed : Test Automation is faster when compared to manual testing.
 Reliable : Automated tests can perform same operation precisely each time.
 Repeatable: Tests can be repeated easily.
 Coverage: Test Automation increase coverage.
 Reusable : Test can be reused in different application versions.

Risks:

Incompatibility of the automation testing tools with the test environment. • Project budget issues. • Vendor issues:
 Overlooking the capability of human testers in specific testing scenarios. • Incorrect evaluation of time and effort required

[Note: Similar answer to the above advantages and risks]

- 3) (a) XPath is one of the most popular html element locators in Test Automation. Briefly explain Absolute and Relative XPath with an example for each. (04 marks)

Absolute XPath:

The complete path to an element on a web page starting from the root node

Example: (starts with single forward slashes)

/html/body/div[1]/section/div[1]/div/div/div/div[1]/div/div/div/div/div[3]/div[1]/div/h4[1]/b

Relative XPath:

A shorter, more flexible path to an element on a web page that starts from an element in the DOM, rather than from the root node.

Example: (starts with double forward slashes)

//*[@class='featured-box']//*[text()='Testing']

- (b) Write down three (03) alternative techniques to locate elements in an HTML document other than XPath. (03 marks)

CSS selectors
JQuery selector
Selenium locators
DOM traversal

[Note: Any valid selector will be considered correct]

(c) Write the correct relative XPath for the given scenarios.

(2x5 = 10 marks)

i) Write XPath to identify 'img' node contains 'src' attribute value as 'toyota'.

```

```

```
//img[contains(@src, 'toyota')]
```

ii) Identify 'select' node using both 'class' and 'name' attributes.

```
+ <select class="htext round" name="vtype" style="background-color: rgb(255, 255, 255);">
  <p>Make</p>
+ <select class="htext round" name="make" style="background-color: rgb(255, 255, 255);">
```

```
//select[@class= 'htext round' and @name='vtype']
```

iii) Write the **absolute** path to expanded 'td' node in the following table.

```
- <table>
  - <tbody>
    + <tr>
    - <tr>
    → <td>URL Categories: Social Networking </td>
    </tr>
    + <tr>
    + <tr>
  </tbody>
</table>
```

```
/html/body/table/tbody/tr[2]/td
```

- iv) Write XPath to identify element using 'id' attribute.

```
<input id="model" class="htext round" type="text" name="model">
```

```
//select[@id= 'model']
```

- v) Write XPath to capture 'h2' node using *text()* function.

```
<div class="homebox">
  <h2>Search Vehicles</h2>
```

```
//h2[text()='Search Vehicles']
```

(d)

Name two (2) main types of mobile applications and briefly explain each type with an example. (8 marks)

Hybrid mobile apps

Native mobile apps

Web based mobile apps

Transpiled apps (Cross platform apps Eg Flutter)

[Note: Student needs to explain two of the above]

- 4) (a) TestNG is a testing framework for Java programming language. Name two (2) unit testing frameworks that were mainly used to develop TestNG framework. (2 marks)

Junit and NUnit

- (b) Write down the output of the following TestNG programme in the given box. Note: Use your knowledge of TestNG annotations execution sequence. (10 marks)

```
public class it6206 {

    @Test
    public void testCase1() { System.out.println ("Test One"); }

    @Test(priority=1)
    public void testCase2() { System.out.println ("Test Two"); }

    @BeforeMethod
    public void beforeMethod() { System.out.println("A"); }

    @AfterMethod
    public void afterMethod() { System.out.println("A"); }

    @BeforeTest
    public void beforeTest() { System.out.println("B"); }

    @AfterTest
    public void afterTest() { System.out.println("B"); }

    @BeforeSuite
    public void beforeSuite() { System.out.println("C"); }

    @AfterSuite
    public void afterSuite() { System.out.println("C"); }

}
```

Write the output here

C
B
A
Test One
A
A
Test Two
A
B
C

- (c) There are different types of testing frameworks available for efficiently test software applications. Write down the correct type of testing framework based on the definitions given below. (1x4 = 04 marks)

- i) Combination of existing testing frameworks to take advantage of the benefits of the different frameworks and mitigate their weaknesses.

Hybrid

- ii) Test execution is basically a sequence of steps defined by keywords and keywords are mapped to objects and actions in a shared object repository.

Keyword driven

- iii) Focus is on executing a test with different input data. Data is read from databases or spreadsheets. There's a clear separation of test scripts and input data.

Data driven

- iv) Tests are independent of one another, often created by recording manual actions and replaying them.

Linear

- (d) List down five (5) criteria that can be used for choosing the right test reporting tool. (05 marks)

User interface

- Clean and easy to use

Usability

- How simple is it to learn and master
- Flexibility to be compatible with a wide range of automation tools

Functionality and features

- Ability to help team members to provide comprehensive and relevant reports

Value

- Reasonable price in relation to the features, capabilities and use case.

[Note: Similar answer will be considered as correct]

- (f) Briefly explain Continuous Integration (CI) and Continuous Delivery (CD). (04 marks)

Continuous Integration:

Continuous Integration (CI) is a method of continuously verifying the condition of a codebase through automated testing.
It is best achieved by integrating CI with version control.

Continuous Delivery:

Approach to regularly deploying artifacts that successfully pass the CI phase to ensure confident around the deployment.

[Note: Similar answer will be considered as correct]
